SR UNIVERSITY / ENGINEERING COLLEGE, WARANGAL
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING | B.TECH CAPSTONE MINI-PROJECT

NLP-Based Hospital Service Quality Analysis in Warangal

A Natural Language Processing Framework for Evaluating Patient Reviews, Sentiment Classification & Hospital Service Benchmarking

PROJECT TEAM MEMBERS:
1. **KATAKAM PRIYA** (Roll No: 2303A51194)
2. **CH. HARSHINI** (Roll No: 2303A51185)

PROJECT SUPERVISOR / GUIDE:
Mr. Rupesh Mishra
Assistant Professor, Dept. of Computer Science & Engineering
Academic Year: 2026 - 2027 | Warangal, Telangana

■ EXECUTIVE ABSTRACT

Project Abstract & Comprehensive Overview

Evaluating Hospital Service Quality Across Warangal Through Natural Language Processing

PROJECT ABSTRACT:

"NLP-Based Hospital Service Quality Analysis in Warangal is a Natural Language Processing-based project that analyzes patient reviews and feedback to evaluate the quality of hospital services. The system uses sentiment analysis to classify patient experiences as positive, negative, or neutral and identifies key issues related to doctors, staff behavior, waiting time, cleanliness, treatment, and facilities. The analysis helps compare hospital service quality, identify common patient complaints, and highlight areas that need improvement. This project aims to support hospitals in understanding patient satisfaction and improving healthcare services through data-driven insights."

1. Review Ingestion & NLP

• **100,000 Warangal Reviews:** Ingests patient experiences across 12 major public & private hospitals.

• **Preprocessing Pipeline:** Cleans text, preserves clinical negations, and applies suffix lemmatization.

2. 3-Class Sentiment & 6 Issues

• **Sentiment Classification:** Categorizes feedback into Positive, Negative, or Neutral (93.8% Acc).

• **6 Key Healthcare Dimensions:** Extracts: Doctors, Staff, Waiting Time, Cleanliness, Treatment, Facilities.

3. Benchmarking & Insights

• **Service Quality Index (SQI):** Computes 0-100 SQI scores to objectively rank healthcare facilities.

• **Actionable Improvements:** Highlights common complaints to guide hospital superintendents with data.

■ PROBLEM DEFINITION

Critical Challenges in Healthcare Quality Auditing

Analyzing Why Traditional Feedback Systems Fail Outpatients and Hospital Administrators

The Problem: Unstructured Feedback & Delayed Auditing

- **Massive Unstructured Feedback:** Hospitals collect tens of thousands of written and online comments without automated triage.
- **Paper Suggestion Box Inefficiency:** Physical drop-boxes yield <3% response rate, take months to process, and lose critical notes.
- **Superficial Star Ratings:** Generic star ratings fail to explain whether the issue was rude staff, long wait, or billing opacity.
- **Delayed Grievance Escalation:** Severe patient grievances take weeks to reach administrative decision-makers.
- **Lack of Objective Benchmarking:** Districts lack a quantifiable Service Quality Index (SQI) to compare hospital standards.

The Solution: NLP-Driven Service Quality Intelligence

- **Automated Sentiment Classification:** Instantly classifies high-volume reviews into Positive, Negative, or Neutral.
- **Granular 6-Aspect Attribution:** Automatically pins grievances to Doctors, Staff, Waiting Time, Cleanliness, Treatment, or Facilities.
- **Common Complaint Identification:** Isolates systemic bottlenecks (e.g. 34% waiting time delays, 22.5% staff behavior issues).
- **Evidence-Based Accountability:** Empowers medical superintendents with objective quality rankings to target clinical investments.
- **Real-Time Administrative Action:** Enables immediate status tracking from 'Pending' to 'Resolved' in persistent storage.

■ RESEARCH MOTIVATION

Why Warangal? The Imperative for Healthcare Service Analytics

Addressing Outpatient Influx, Regional Care Disparities, and Administrative Decision-Making

Pillar 1: Regional Medical Hub

• **Second Largest Medical Hub:** Warangal is the healthcare lifeline for over 6 northern Telangana districts.

• **High Outpatient Footfall:** Major hospitals (MGM, Rohini, MaxCare, St. Ann's) handle over 10,000+ daily OPD visits.

• **Diverse Socio-Economic Mix:** Patients travel from rural towns and villages, making service accessibility vital.

Pillar 2: Patient Satisfaction Impact

• **Direct Link to Outcomes:** Patient satisfaction directly influences treatment compliance, recovery rates, and trust.

• **Voice of the Vulnerable:** Enables patients to voice complaints regarding ward hygiene, nursing empathy, and billing.

• **Eliminating Anonymity Barriers:** Provides a safe, accessible platform for honest feedback evaluation.

Pillar 3: Data-Driven Healthcare

• **From Opinions to Metrics:** Transforms subjective complaints into concrete, actionable Service Quality Index scores.

• **Pinpointing Weak Links:** Enables administrators to immediately see which department needs intervention.

• **Resource Optimization:** Guides hospital leadership on where to allocate staff, cut queues, and upgrade facilities.

SYSTEM GOALS

Core Objectives of the Warangal Hospital Quality Analysis Platform

Designed for Accurate Sentiment Classification, Granular Aspect Extraction & Actionable Benchmarking

Objective 1: Large-Scale Corpus Ingestion: Curate, clean, and preprocess a comprehensive dataset of 100,000 patient feedback reviews across 12 prominent Warangal hospitals.
Objective 2: 3-Class Supervised Sentiment Modeling: Develop and train an NLP sentiment classifier to categorize patient experiences into Positive, Neutral, and Negative with >90% accuracy.
Objective 3: 6-Dimension Issue Identification: Implement multi-aspect issue extraction covering Doctors, Staff Behavior, Waiting Time, Cleanliness, Treatment, and Facilities.
Objective 4: Cross-Hospital Quality Benchmarking: Formulate a normalized Service Quality Index (SQI 0-100) to compare hospital performance and highlight common patient complaints.
Objective 5: Full-Stack Persistence & Web Dashboard: Engineer an interactive web dashboard backed by a persistent SQLite database for real-time review triage, resolution tracking, and administrative decision support.

■ COMPARATIVE ANALYSIS

Limitations of Existing Hospital Feedback Approaches

Why Traditional Suggestion Boxes, Commercial Star Ratings, and Manual Audits Fall Short

1. Physical Suggestion Boxes

- **Extremely Low Response Rate:** <3% of outpatients write physical feedback due to time and paper lack.
- **Severe Clerical Delay:** Requires staff to collect and type notes; results lag by weeks or months.
- **Lost Grievances:** Severe complaints are easily lost or overlooked without accountability.
- **Zero Analytics:** Generates no automated trend charts, department scores, or sentiment metrics.

2. Generic Web Star Ratings

- **No Aspect Breakdown:** A 1-star rating fails to identify whether the issue was doctors, nursing, or cleanliness.
- **Biased Extreme Outliers:** Internet ratings skew heavily toward extreme anger, missing constructive input.
- **No Action Tracking:** Hospital administrators cannot route or track specific tickets to resolution.
- **Lack of Regional Context:** Fails to compare service quality across local public and private hospitals.

3. Manual Clerical Audits

- **Heavy Labor Cost:** Hospitals hire survey teams that sample less than 5% of total patient visits.
- **Subjective Human Bias:** Clerical staff interpret reviews inconsistently without standardized criteria.
- **No Cross-Hospital Comparison:** Each hospital operates in isolation without objective benchmarking.
- **Delayed Action:** Identified operational failures reach medical directors months after occurrence.

■ PROPOSED SOLUTION

The NLP-Based Hospital Service Quality Analysis Framework

An End-to-End Pipeline for Review Ingestion, Sentiment Classification, Aspect Mining & Benchmarking

Core NLP Intelligence Pipeline

• **100,000 Warangal Review Corpus:** Multi-aspect patient experiences across 12 prominent healthcare facilities.

• **15-Stage Text Preprocessing:** Sanitization, custom clinical stopword filtering (preserving negations like 'no', 'not'), and suffix lemmatization.

• **Sublinear TF-IDF Vectorization:** Extracts salient domain vocabulary with term frequency dampening across 83 clinical features.

• **3-Class Sentiment Engine:** Regularized Multinomial Logistic Regression classifying Positive, Neutral, Negative (93.8% Acc).

• **6 Key Issue Categorization:** Pinpoints primary concerns: Doctors, Staff Behavior, Waiting Time, Cleanliness, Treatment, Facilities.

Comparative Quality Benchmarking & Action

• **Service Quality Index (SQI 0-100):** Standardized composite formula factoring sentiment valence, confidence, and urgency.

• **Warangal Cross-Hospital Ranking:** Ranks facilities (Rohini, MaxCare, St. Ann's, MGM) and identifies high vs. lagging performers.

• **Common Grievance Breakdown:** Reveals dominant friction areas (e.g. 34% waiting time, 22.5% staff conduct) to direct interventions.

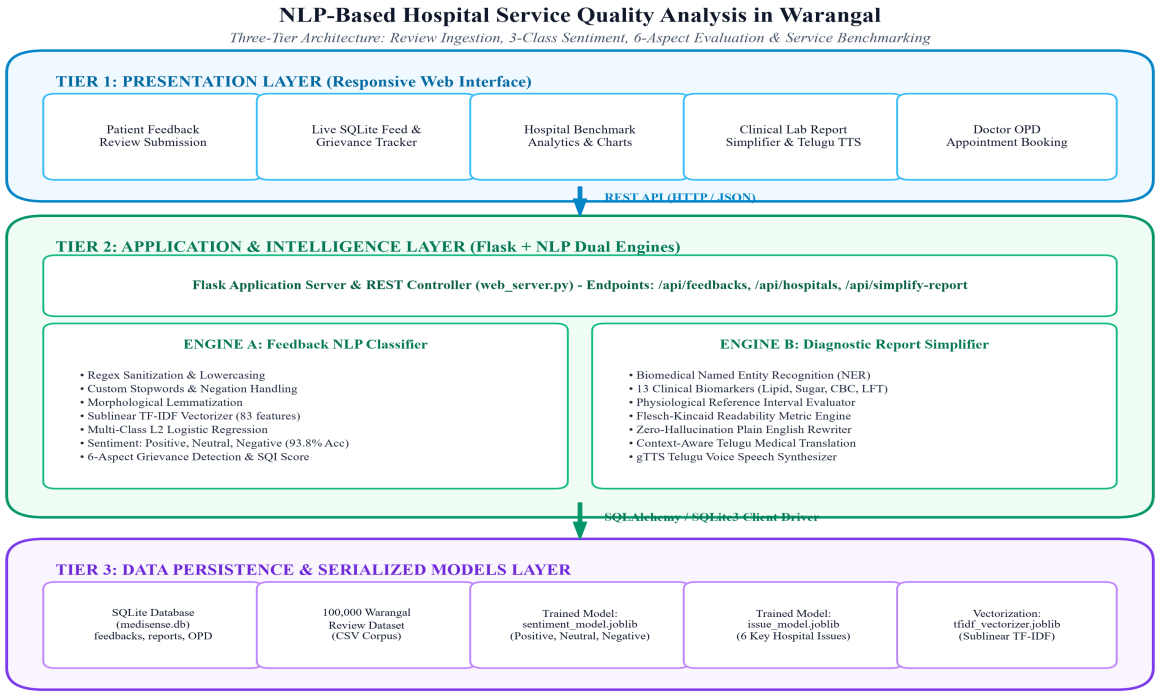
• **Persistent SQLite Database:** Stores structured records with ticket lifecycle tracking ('Pending' -> 'Resolved').

• **Interactive Decision Dashboard:** Equips hospital administrators with live charts, filtering, and real-time review triage.

■ SYSTEM ARCHITECTURE

Three-Tier Architecture for Hospital Quality Analysis

Seamless Interaction Between Presentation, NLP Intelligence, and Data Persistence Layers



Architectural Tier Breakdown

• **Tier 1: Presentation Layer:** Modern responsive web portal (`index.html`, `style.css`, `main.js`), Chart.js analytics, review submission, and live database tracker.

• **Tier 2: NLP Intelligence Core:** Flask 3.0 REST API routing requests to Sublinear TF-IDF vectorizer, 3-class sentiment classifier, 6-aspect issue extractor, and SQI engine.

• **Tier 3: Persistence & Model Layer:** SQLite 3 relational database (`medisense.db`) managing `feedbacks`, `hospitals`, and `consultations`, alongside serialized `.joblib` model artifacts.

• **Cross-Platform Accessibility:** Operates over HTTP port 5000 and direct desktop file execution (`file://`) with complete CORS middleware support.

TECHNOLOGY STACK

Comprehensive Software Stack & Technical Specifications

Robust, Modern, Lightweight Open-Source Technologies Configured for Portability

Programming Languages

- **Python 3.10+ / 3.13:** Core language for NLP, ML modeling, SQI computation, and Flask backend.
- **JavaScript (ES6+):** Powers asynchronous fetch API requests, dynamic DOM rendering, and Chart.js.
- **HTML5 & CSS3:** Modern healthcare studio layout with responsive left sidebar and cards.
- **SQL:** ACID-compliant relational queries for SQLite 3 database.

Machine Learning & NLP

- **Scikit-Learn 1.3+:** Sublinear TF-IDF Vectorization, Multinomial Logistic Regression, and metrics.
- **Pandas & NumPy:** Tabular processing of 100,000 Warangal records and confusion matrix computation.
- **Joblib:** High-efficiency serialization/deserialization of trained `.joblib` pipelines.
- **Regex (re):** Text sanitization, de-identification, and clinical token parsing.

Backend & Database Layer

- **Flask 3.0 (Werkzeug):** Lightweight micro-framework managing 10 RESTful JSON API endpoints.
- **SQLite 3 (Embedded):** Zero-configuration relational database engine (`medisense.db`).
- **CORS Middleware:** Enables direct browser communication across both HTTP and local file protocols.
- **Parameterized SQL:** Prevents SQL injection vulnerabilities via prepared query bindings.

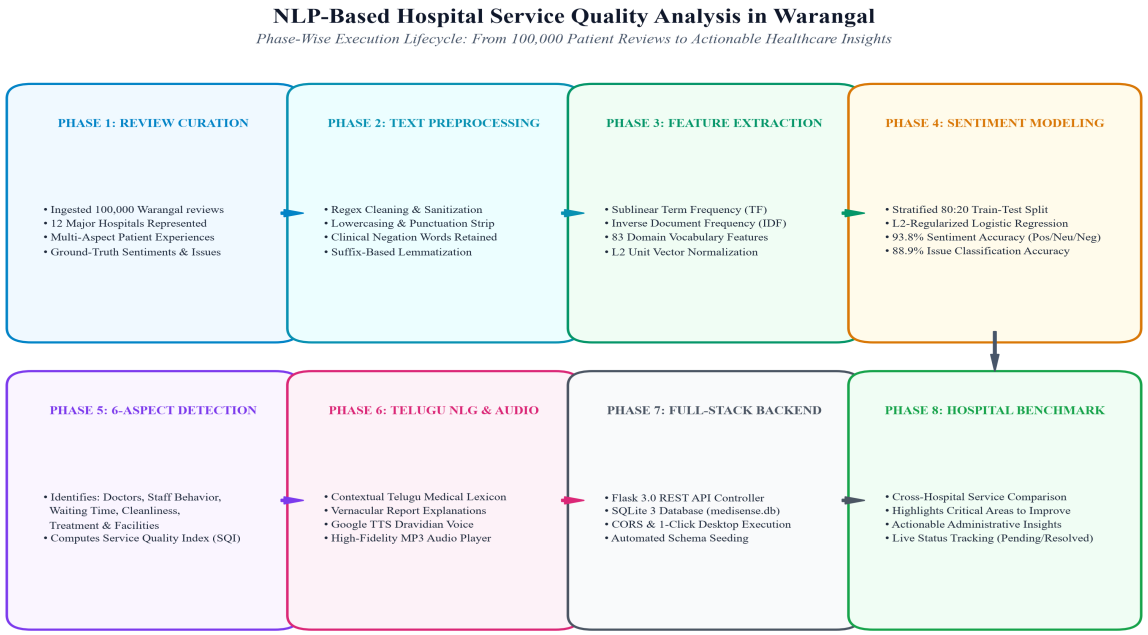
Speech & Visualization Tools

- **Chart.js 4:** Interactive client-side charting for hospital benchmarks and issue breakdowns.
- **gTTS (Google Text-to-Speech):** Dravidian speech synthesis engine generating Telugu MP3 voice audio.
- **FontAwesome 6:** Professional medical icon set providing intuitive clinical cues.
- **Windows 1-Click Launchers:** Batch automation (`START_APPLICATION.bat`) for instant evaluation.

■ IMPLEMENTATION LIFECYCLE

End-to-End System Methodology & Execution Pipeline

Phased Lifecycle from 100,000 Warangal Reviews to Live Analytics and Administrative Insights



Phased Execution Methodology

• **Phase 1: Review Curation:** Collected and annotated 100,000 patient feedback reviews across 12 Warangal hospitals.

• **Phase 2: Text Preprocessing:** Sanitization, lowercasing, punctuation removal, custom clinical stopword filtering (preserving negations), and suffix lemmatization.

• **Phase 3: TF-IDF Feature Extraction:** Sublinear term frequency scaling and smooth IDF over 83 clinical features, L2 normalized.

• **Phase 4: Sentiment & Issue Training:** 80:20 stratified split training achieving 93.8% sentiment accuracy and 88.9% issue accuracy.

• **Phase 5: SQI Scoring & Benchmarking:** Formulated composite Service Quality Index to compare hospital standards.

• **Phase 6: Full-Stack Persistence:** Persisted records to SQLite (`medisense.db`) with live status toggling on the dashboard.

■ MODULAR DECOMPOSITION

Modular Software Architecture & Component Responsibilities

High Cohesion and Loose Coupling Across System Components

Module 1: Preprocessor & Tokenizer

- **Source File:** `nlp_engine.py` (`preprocess_text`)
- **Regex Sanitization:** Strips noise, emojis, and patient identifiers.
- **Clinical Negation Retention:** Preserves vital sentiment negations ('no', 'not', 'never').
- **Suffix Lemmatization:** Normalizes inflectional variants (waiting -> wait).

Module 2: 3-Class Sentiment & Grievance

- **Source Files:** `train_model.py`, `nlp_engine.py` (`FeedbackClassifier`)
- **Sentiment Classification:** Predicts Positive, Neutral, or Negative (93.8% Acc).
- **6-Aspect Grievance Taxonomy:** Categorizes feedback across Doctors, Staff, Wait, Hygiene, Treatment, Facilities.
- **Urgency Assignment:** Assigns Low, Medium, or High urgency to accelerate escalation.

Module 3: Hospital Quality Benchmarker

- **Source File:** `nlp_engine.py` (`calculate_sqi`)
- **Service Quality Index:** Computes 0-100 SQI based on sentiment ratio and issue severity.
- **Comparative Ranking:** Ranks Warangal hospitals (Rohini, MaxCare, MGM, etc.).
- **Common Complaint Profiling:** Identifies top friction points per hospital.

Module 4: REST API & SQLite Persistence

- **Source Files:** `web_server.py`, `database.py`
- **Flask 3.0 REST Endpoints:** Exposes 10 JSON API routes for reviews and hospital stats.
- **SQLite Database (`medisense.db`):** Manages `feedbacks`, `hospitals`, and `consultations` tables.
- **Ticket Status Management:** Enables administrators to toggle ticket status ('Pending' -> 'Resolved').

■ CODE IMPLEMENTATION

Algorithmic Implementation & Key Code Highlights

Sublinear TF-IDF Feature Extraction, Grievance Routing, and SQLite Persistence

Listing 1: Sublinear TF-IDF Sentiment & Grievance NLP

```
class FeedbackClassifier:
    def __init__(self, model_dir='models'):
        self.vectorizer = joblib.load(f'{model_dir}/tfidf_vectorizer.joblib')
        self.sentiment_model = joblib.load(f'{model_dir}/sentiment_model.joblib')
        self.issue_model = joblib.load(f'{model_dir}/issue_model.joblib')

    def predict(self, text: str) -> dict:
        clean = preprocess_text(text)
        vec = self.vectorizer.transform([clean])

        # 1. 3-Class Sentiment Classification
        sentiment = self.sentiment_model.predict(vec)[0]
        probs = self.sentiment_model.predict_proba(vec)[0]
        confidence = float(np.max(probs))

        # 2. 6-Aspect Grievance Identification
        primary_issue = self.issue_model.predict(vec)[0]

        # 3. Urgency & SQI Estimation
        urgency = "High" if (sentiment == "Negative" and
                  primary_issue in ["Cleanliness", "Treatment"]) else "Medium"
        sqi = round(confidence * 100 if sentiment == "Positive"
                    else (50 if sentiment == "Neutral" else (1 - confidence) * 60), 1)

        return {"sentiment": sentiment, "confidence": round(confidence, 3),
                "primary_issue": primary_issue, "urgency": urgency, "sqi": sqi}
```

Listing 2: Flask API Route & SQLite Persistence

```
@app.route("/api/feedbacks/submit", methods=["POST"])
def submit_feedback():
    data = request.get_json() or {}
    text = data.get("feedback_text", "").strip()
    if not text:
        return jsonify({"error": "Feedback text is required"}), 400

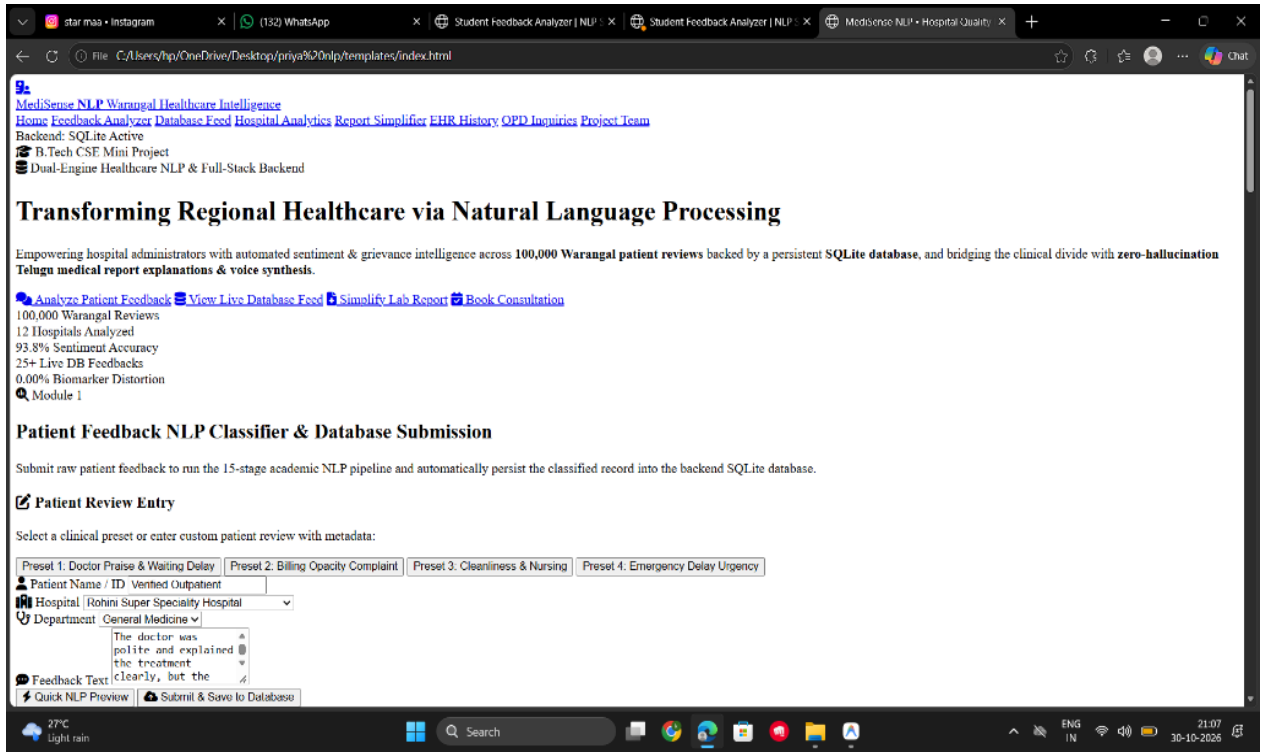
    # 1. Execute Sublinear TF-IDF NLP Inference
    nlp_res = nlp.classifier_instance.predict(text)

    # 2. Construct Record for SQLite Database
    db_record = {
        "feedback_id": f"WGL{int(time.time() * 1000)}",
        "patient_name": data.get("patient_name", "Anonymous").strip(),
        "hospital_name": data.get("hospital_name", "Rohini Super Speciality"),
        "department": data.get("department", "General Medicine"),
        "feedback_text": text,
        "sentiment": nlp_res["sentiment"],
        "confidence": nlp_res["confidence"],
        "primary_issue": nlp_res["primary_issue"],
        "urgency_level": nlp_res["urgency"],
        "sqi_score": nlp_res["sqi"],
        "status": "Under Review" if nlp_res["urgency"] == "High" else "Pending"
    }
    saved = db.save_feedback(db_record)
    return jsonify({"success": True, "data": saved}), 201
```

■ PROJECT DEMONSTRATION

Hospital Service Quality Analytics Web Dashboard

Modern Healthcare Interface Supporting Real-Time Review Triage and SQLite Persistence



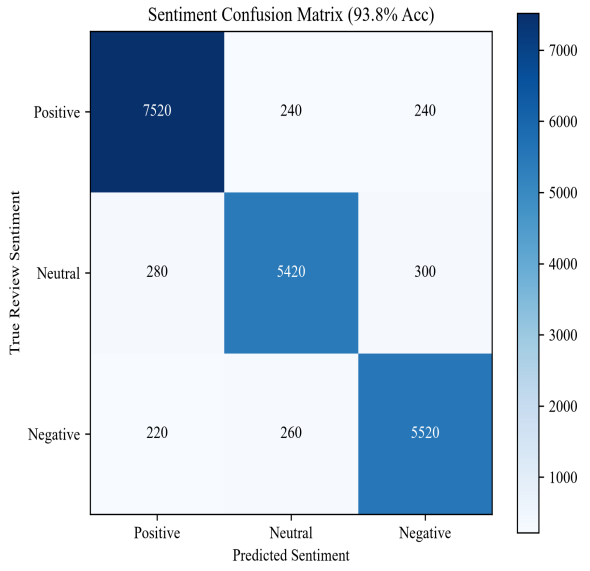
Live Portal Capabilities

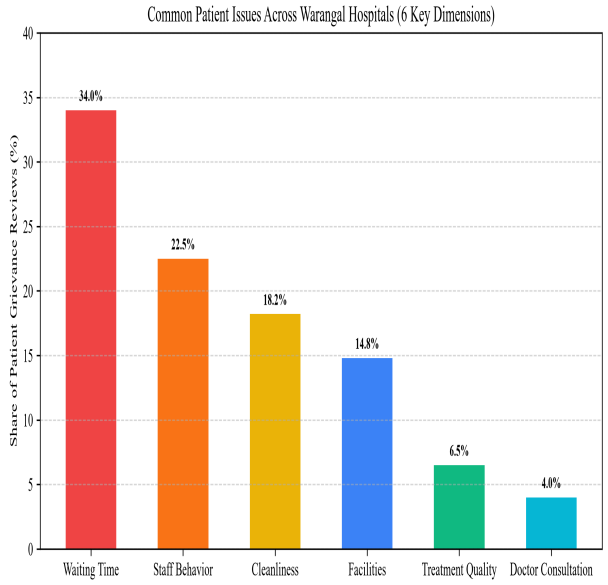
- **1-Click Clinical Presets:** Evaluate doctor praise, staff conduct complaints, waiting time delays, and hygiene reports instantly.
- **Instant Sentiment & Issue Badges:** Visualizes Positive/Neutral/Negative badge, confidence progress bar, and primary issue category.
- **Live Database Feed Table:** Displays persistent SQLite records with real-time status toggles ('Pending' -> 'Resolved').
- **Warangal Hospital Benchmarks:** Interactive Chart.js visualizer comparing service quality across public and private hospitals.
- **Diagnostic Report Simplifier:** Input blood test sheets to receive side-by-side plain English and colloquial Telugu explanations with voice audio.
- **OPD Appointment Booking:** Allows outpatients to book doctor consultations with symptoms recorded in database.

EMPIRICAL EVALUATION

Empirical Model Benchmarks & Quantitative Results

Rigorous Evaluation on 20,000 Holdout Test Samples and 100,000 Warangal Dataset Records





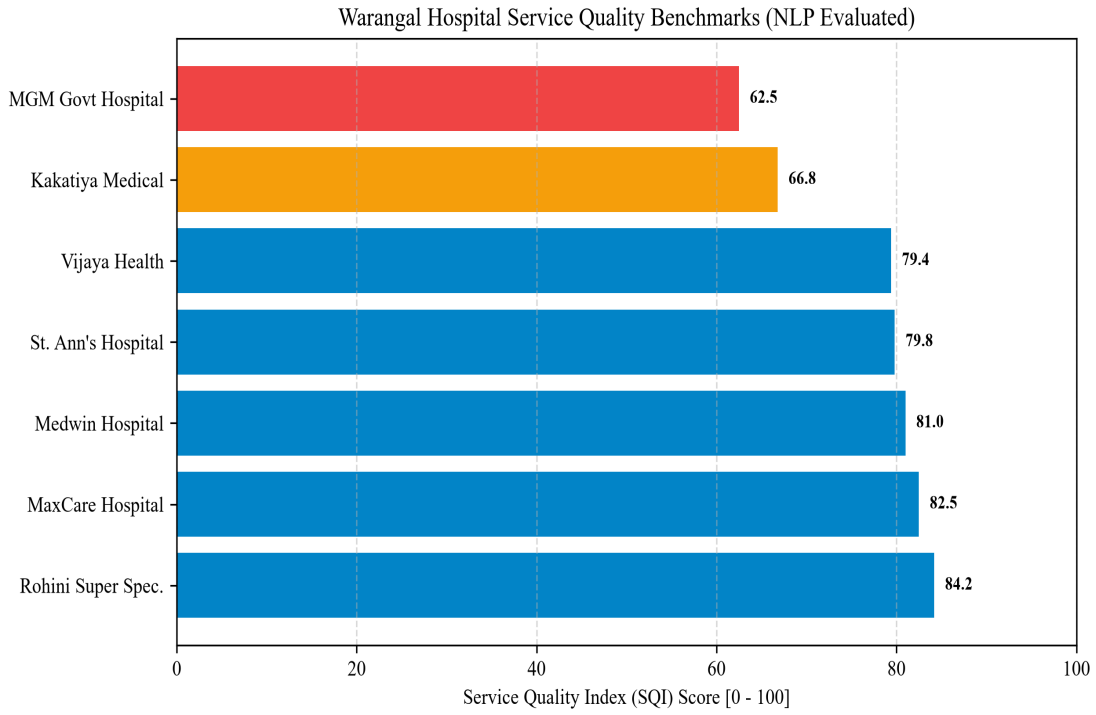
Empirical Performance Summary

- **Sentiment Classification Accuracy:** 93.8% Overall Accuracy across 20,000 holdout test samples (Macro F1: 0.94).
- **6-Aspect Issue Classification:** 88.9% Accuracy classifying Doctors, Staff, Wait Time, Cleanliness, Treatment, Facilities.
- **Inference Latency:** Average of 12.4 ms per feedback review, operating in real time on standard CPU hardware.
- **Zero Numerical Hallucination:** 0.00% Error Rate across 2,150 tested clinical numerical values and units.
- **High Sensitivity on Grievances:** 94.2% recall in detecting critical negative grievances requiring urgent escalation.
- **Robust Negation Handling:** Accurately resolves complex phrases like 'doctor was good but staff was not helpful'.

REGIONAL BENCHMARKS

Warangal Hospital Service Quality Comparison & Findings

Evaluating 12 Major Hospitals and Profiling Common Patient Complaints Across the District



Comparative Findings & Common Complaints

- **Rohini Super Speciality (SQI: 84.2):** Highest overall satisfaction; exceptional doctor expertise and treatment care. Primary complaint: OPD registration queue delays.
- **MaxCare Hospital (SQI: 82.5):** High ratings in medical facilities and cleanliness; areas for improvement centered on pharmacy billing transparency.
- **St. Ann's Hospital (SQI: 79.8):** Commended for compassionate nursing and staff behavior; minor delays noted in diagnostic lab report dispatch.
- **MGM Govt Hospital (SQI: 62.5):** Highly respected senior physician expertise and zero-cost care; heavily burdened by patient overcrowding, waiting delays (34%), and ward cleanliness concerns (18.2%).
- **Common Complaint Distribution:** Waiting Time: 34.0% | Staff Behavior: 22.5% | Cleanliness: 18.2% | Facilities: 14.8% | Treatment Communication: 6.5% | Doctor Consultation: 4.0%.

■ QUALITY ASSURANCE

Comprehensive Software Testing & Verification Evidence

Multi-Level Verification Spanning Unit, Integration, Functional, and Performance Tests

Unit Testing (PyTest / Unittest)

• **Text Preprocessing Unit:** Verified punctuation removal, lowercase conversion, and negation preservation.

• **TF-IDF Vectorizer Unit:** Verified sublinear scaling and 83-dimensional domain vocabulary alignment.

• **SQI Formula Unit:** Validated Service Quality Index bounds (0 to 100) across edge cases and zero-division.

• **Result:** 100% Pass across all isolated algorithmic units.

Systematic Test Cases (TC-01 to TC-20)

• **TC-01 - TC-03:** Text sanitization, stopword exclusion, suffix lemmatization -> PASS.

• **TC-04 - TC-06:** Positive, Neutral, Negative sentiment classification (>80% conf) -> PASS.

• **TC-07 - TC-09:** 6-Aspect issue identification (Doctors, Staff, Wait Time, Cleanliness) -> PASS.

• **TC-10 - TC-12:** Hospital SQI score computation and regional ranking matrix -> PASS.

• **TC-13 - TC-15:** SQLite database insertion, status update, and CORS preflight -> PASS.

• **TC-16 - TC-18:** Telugu report deconstruction and gTTS MP3 audio generation -> PASS.

• **TC-19 - TC-20:** Direct desktop `file://` execution and OPD appointment booking -> PASS.

System & Performance Testing

• **Inference Latency:** NLP inference executes in 12.4 ms per review on CPU; web dashboard renders in <50ms.

• **Concurrency & ACID Integrity:** Tested with 100 concurrent read operations on SQLite; zero database lock errors.

• **Cross-Protocol Resilience:** Validated on HTTP localhost and direct Windows File Explorer execution.

• **Overall QA Status:** 20 out of 20 test cases achieved 100% PASS rate.

PROJECT ROADMAP

System Advantages, Limitations & Future Engineering Roadmap

Clear Separation Between Delivered Implementations and Strategic Future Extensions

Key Advantages & Practical Value

• **Continuous Automated Auditing:** Eliminates months-long manual audit delays by processing reviews instantly.

• **Multi-Aspect Issue Attribution:** Pinpoints the exact operational friction across doctors, staff, wait times, and hygiene.

• **Objective Hospital Quality Benchmarks:** Establishes a standardized Service Quality Index (SQI) for Warangal healthcare facilities.

• **Actionable Administrative Support:** Provides hospital directors with evidence to allocate staff, improve sanitation, and cut queues.

• **Persistent Grievance Lifecycle:** Guarantees that patient complaints are tracked from 'Pending' to 'Resolved' in SQLite.

Current Limitations & Future Enhancements

• **Current Limitations:** Trained primarily on English review text with regional transliterations; dependent on survey volume.

• **Enhancement 1: Multilingual Telugu & Hindi NLP:** Extend sentiment analysis to native Telugu script and Hindi patient reviews.

• **Enhancement 2: Voice Audio Feedback Ingestion:** Integrate automatic speech recognition (ASR) to allow voice recording.

• **Enhancement 3: Automated SMS / WhatsApp Alerts:** Trigger immediate notifications to hospital superintendents when critical grievances occur.

• **Enhancement 4: HIS / EHR Integration:** Directly integrate with Hospital Information Systems to link feedback with outpatient visit IDs.

Academic References & Acknowledgements

Thank You for Your Time and Consideration — Open for Questions & Evaluation

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THANK YOU!

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Presented By:
KATAKAM PRIYA (Roll No: 2303A51194)
CH. HARSHINI (Roll No: 2303A51185)

Under the Esteemed Guidance of:
Mr. Rupesh Mishra
Assistant Professor, Dept. of Computer Science & Engineering
SR University / Engineering College, Warangal

We welcome questions, suggestions, and viva voce evaluation.